

COMPETITIVE INTELLIGENCE REPORT

Pentixapharm in CXCR4-Targeted Theranostics

First-in-Class Radiopharmaceutical Platform — Pipeline Analysis and
Competitive Landscape

Prepared for:

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27 May 2026 | Sources: ClinicalTrials.gov, EMA, PubMed, Yahoo Finance, Company Website

Disclaimer & Methodology

Purpose: Independent competitive intelligence analysis of the CXCR4-targeted radiopharmaceutical landscape, with focus on Pentixapharm's theranostic platform (PentixaFor/PentixaTher).

Data Sources:

- **ClinicalTrials.gov** — All trial statuses verified live on 27 May 2026
- **EMA Human Medicines Register** — EU approval status for competitor therapies
- **PubMed/MEDLINE** — Scientific publications (PMID cited)
- **Pentixapharm Company Website** — Pipeline, management, news (accessed 27 May 2026)
- **Open Targets (EMBL-EBI)** — Drug-target relationships
- **Yahoo Finance** — Market data (PTP.DE, XETRA, 27 May 2026)

Company press releases reviewed through 27 May 2026; key milestones cited include: PentixaTher advancement to Dose Level 4 in AML (Sept 2025), PANDA Phase 3 IND submission to FDA (May 2026), dual FDA "Study May Proceed" letters for hemato-oncology INDs (Feb 2026), and pipeline prioritisation with preclinical halt (Oct 2025).

FAERS analysis: Not conducted — [⁹⁰Y]Y-PentixaTher is an investigational radiopharmaceutical with no FDA-approved NDC. FAERS data for investigational drugs produces false positives due to NDC overlap and is excluded from this analysis.

Enrollment notation: All enrollment figures from ClinicalTrials.gov reflect **target enrollment** (planned) unless explicitly marked as "disclosed actual" with a date.

Financial data note: Pentixapharm Holding AG was listed on 3 October 2024 as a spin-off from Eckert & Ziegler Group. As a recently listed small cap with limited analyst coverage, some financial metrics (shares outstanding, cash position) were not available via public market data APIs at the time of writing and are noted where applicable.

Disclaimer: This is an independent analysis prepared from publicly available data. It does not constitute investment advice, medical advice, or regulatory guidance. The author has no financial interest in Pentixapharm Holding AG.

1. Executive Summary

Pentixapharm Holding AG (PTP:GR) is a clinical-stage radiopharmaceutical company headquartered in Berlin, Germany, developing a first-in-class CXCR4 theranostic platform. Founded in 2019, the company was listed on the Frankfurt Stock Exchange (Prime Standard) in October 2024 as a spin-off from the Eckert & Ziegler Group. The company is led by Dr. Dirk Pleimes (Group CEO & CMO).

Pentixapharm occupies a unique position in the radiopharmaceutical market: the identical CXCR4 ligand serves as both a diagnostic (^{68}Ga -labelled: PentixaFor) and a therapeutic (^{90}Y -/ ^{177}Lu -labelled: PentixaTher) agent — a true theranostic approach. With over 2,000 administered patient doses and more than 150 peer-reviewed publications, the scientific foundation of the CXCR4 target is extensively validated.

Central Theses of This Report:

- 1. Unoccupied orphan space:** No EMA- or FDA-approved therapy exists specifically for PCNSL — PentixaTher could be first-in-class
- 2. Pipeline-in-a-pipeline:** Five indications from one ligand scaffold — the diagnostic Phase 3 in hypertension de-risks the entire therapeutic pipeline
- 3. M&A catalyst:** Radiopharma M&A (BMS-Rayzebio \$4.1B, Eli Lilly-Point \$1.4B, Novartis-AAA \$3.9B) creates a defined exit pathway for platform companies

Key Metrics	Value
Stock Price (27 May 2026)	€2.27 (XETRA)
52-Week Range	€1.30 – €3.70
Therapeutic Pipeline	2 Ph3-ready (diagnostic) + 2 Ph1/2 (therapeutic) + 1 preclinical
Lead Therapeutic Asset	^{90}Y -PentixaTher — CXCR4 endoradiotherapy in CNS Lymphoma (Ph1/2) & AML
Diagnostic Lead	^{68}Ga -PentixaFor — Phase 3 PANDA in Hypertension, IND May 2026
Next Catalyst	PANDA Ph3 Start H2 2026, AML Dose Level 4 data 2026
Scientific Foundation	>2,000 patients, >150 publications, 20+ investigator-initiated studies

2. Company Overview

2.1 Corporate Structure

Pentixapharm Holding AG is the publicly traded parent company (PTP:GR, Frankfurt Prime Standard). The operating subsidiary Pentixapharm AG, based in Berlin and Würzburg, employs approximately 35 people with a high R&D share (>80%). The company was founded in 2019 as a spin-off from the Eckert & Ziegler Group — an established radiopharmaceutical company with decades of experience in radioisotope production and supply chain.

2.2 Management

Name	Position	Background
Dr. Dirk Pleimes	Group CEO & CMO	Bayer, Novartis — radiopharma & oncology experience
Henner Kollenberg	CBO	Business development, licensing negotiations
Dr. Erik Merten	CTO (since March 2026)	Technology development, radiopharmaceuticals
Dr. Andreas Eckert	Chairman of Supervisory Board	Former CEO Eckert & Ziegler, radiopharma pioneer

2.3 Financial Situation

Financial Metric	Value	Source
Stock Price (27 May 2026)	€2.27	Yahoo Finance (PTP.DE, XETRA)
52-Week High / Low	€3.70 / €1.30	Yahoo Finance
Daily Trading Volume	~3,300 shares (thin)	Yahoo Finance
Market Capitalisation	Est. €50–150M ¹	Limited public data
Cash Position	Not publicly disclosed ²	Eckert & Ziegler spin-off
Loss Forecast	Reduced (Ad-hoc Nov 2025)	Company ad-hoc announcement

¹ Shares outstanding not available via public market data APIs. Freshly listed spin-off (Oct 2024).

² Q1 2026 report published 6 May 2026; detailed cash position not extracted via available data channels.

Cash Runway Assessment: Moderate (6–18 months estimated). The pipeline prioritisation (clinical > preclinical, announced Oct 2025) and the reduced loss forecast (Ad-hoc Nov 2025) indicate disciplined cash management. As a recently listed public company without product revenue, capital market dependency remains. The spin-off from Eckert & Ziegler provided an initial capital base, but the next financing round (equity, partnership, or M&A) will likely be

needed within 12–18 months to fund the PANDA Phase 3 and continued CNS Lymphoma/AML development.

3. Pipeline Deep Dive — PentixaTher

3.1 [⁹⁰Y]Y-PentixaTher in CNS Lymphoma (NCT06132737)

Parameter	Value
Trial	[⁹⁰ Y]Y-PTT Endoradiotherapy in CNS Lymphoma Patients
Phase	1/2
Design	Open-label, single-arm, Best-of-5 dose escalation
Enrollment (target)	15
Start Date	7 Nov 2023
Primary Completion	23 Sep 2027
Final Completion	26 Mar 2028
Primary Endpoint	Safety: (S)AE incidence & severity (NCI CTCAE v5)
Secondary	ORR, PFS, OS at Month 1, 3, 6, 9, 12
Sites	2 (TU Munich/Rechts der Isar + University Hospital Essen)
PI	Prof. Dr. Bastian von Tresckow (Essen)
Indication	R/R Primary CNS Lymphoma (PCNSL) & Secondary CNS Lymphoma (SCNSL)

Timeline Assessment: 15 patients over ~4 years (2023–2027) in a Best-of-5 dose escalation for PCNSL (incidence ~0.5/100,000) with only 2 clinical sites. This is a **conservative but appropriate timeline** for an orphan CNS indication. The parallel AML PentixaTher trial reaching Dose Level 4 (Sept 2025) confirms functional enrollment across the platform.

3.2 [¹⁷⁷Lu]Pentixather in AML — Investigator-Initiated

Pentixather is being investigated in an investigator-initiated study (University Hospital Würzburg) as myeloablative conditioning before allogeneic stem cell transplantation in R/R AML. Dose Level 4 was reached in September 2025 (Pentixapharm news, 25 Sept 2025). A companion publication in *Theranostics* (2025, PMID: 39744224) documents CXCR4-directed endoradiotherapy plus total body irradiation as a conditioning regimen.

Clinical Significance: AML shows the highest CXCR4 overexpression among hematologic malignancies. The conditioning-before-HSCT mechanism is distinct from cytoreductive monotherapy — Pentixather acts as part of the conditioning regimen, a differentiated use case that supports translation into other CXCR4+ indications.

3.3 [⁶⁸Ga]Ga-PentixaFor — Diagnostic Platform

The diagnostic arm is significantly more advanced:

- **Phase 3 PANDA** in hypertension (Prim. Aldosteronism) — IND submitted to FDA May 2026
- **FDA "Study May Proceed"** for dual hemato-oncology INDs (Feb 2026)
- **Phase 2 data** on PentixaFor in Prim. Aldosteronism peer-reviewed and published (Feb 2026)
- **5 Phase 2 studies** completed or ongoing (including multiple investigator-initiated studies in China)
- **20+ investigator-initiated studies (IIS)** generating real-world evidence across indications

4. Therapeutic Area — CNS Lymphoma

4.1 Disease Background

Primary CNS Lymphoma (PCNSL) is a rare (<5/10,000), aggressive non-Hodgkin lymphoma confined to the central nervous system. Incidence is approximately 0.5 per 100,000 person-years. Standard therapy consists of high-dose methotrexate (HD-MTX)-based regimens, often combined with whole-brain radiotherapy — both associated with significant neurotoxicity. **No EMA- or FDA-approved drug exists specifically for PCNSL.**

4.2 Competitive Landscape — CNS Lymphoma

Therapy	Mechanism	Phase	Status
HD-MTX + Rituximab + Chemo	Chemotherapy	Standard (off-label)	Neurotoxicity, no targeted MOA
Tafasitamab + Lenalidomide (NCT05351593)	Anti-CD19 mAb + IMiD	Ph1/2	RECRUITING, PCNSL/SCNSL
Anti-CD19 CAR-NK (NCT06827782)	CAR-NK cell therapy	Ph1	China, ENROLLING
[⁹⁰ Y]Y-PentixaTher	CXCR4 endoradiotherapy	Ph1/2	RECRUITING — first-in-class CXCR4

Competitive Differentiation: PentixaTher is the only CXCR4-targeted approach in CNS lymphoma. The diagnostic pre-screening (PentixaFor PET identifies CXCR4+ lesions) enables patient selection — unlike non-targeted chemotherapies. The documented blood-brain barrier penetration of the CXCR4 ligand is a key differentiator versus antibody-based approaches and larger CAR-T cells.

4.3 AML Competitive Context

CXCR4 expression in AML is functionally significant (homing, niche interaction, resistance). Pentixather as myeloablative conditioning occupies its own niche — not in direct competition with venetoclax, gilteritinib, or IDH inhibitors, but as a conditioning boost before HSCT. Next benchmark: Dose Level 4 data readout expected 2026.

5. Strategic Positioning & Recommendations

5.1 SWOT Analysis

Strengths	Weaknesses
• First-in-class CXCR4 theranostic — fully differentiated from ADC, CAR-T, BiTE • 2,000+ patients, 150+ publications = validated target with extensive real-world data • Diagnostic Phase 3 de-risks therapeutic pipeline • Spin-off from Eckert & Ziegler = deep radiopharma know-how and supply chain	• No clinical efficacy data for PentixaTher monotherapy yet • Small studies (target: 15 CNS Lymphoma) — limited statistical power • Early stage (Ph1/2) — high technical risk • Capital market dependency (no product revenue)
Opportunities	Threats
• Radiopharma M&A wave (4 deals >\$1B since 2023) • Unoccupied PCNSL market — Orphan Drug potential • Single ligand scaffold = multiple indications from one asset • Diagnostic creates commercial clinical foothold	• CXCR4 target not tumour-specific (bone marrow toxicity risk) • Radiopharma logistics (short half-life, supply chain complexity) • Competition from established players (Novartis, Bayer, Eli Lilly) • Slow enrollment in orphan CNS indications

5.2 Strategic Recommendations

1. **CNS Lymphoma as accelerated path:** Upon positive DLT data, priority should be given to an FDA Orphan Drug Designation application for PCNSL. The unoccupied market (~1,800 new diagnoses/year in EU+US) justifies accelerated development.
2. **Use diagnostic as value driver:** The PANDA Phase 3 in hypertension is not merely a diagnostic program — it validates the CXCR4 platform with the FDA. Positive Phase 3 data would increase the entire therapeutic pipeline's value.
3. **AML conditioning as proof-of-concept:** Myeloablative conditioning with Pentixather + TBI (Theranostics 2025) is an unconventional but low-risk entry into AML — lower efficacy hurdle (engraftment rate) compared to monotherapy endpoints (CR/CRi).
4. **M&A readiness:** At radiopharma valuations of 2–4x pipeline value (BMS-Rayzebio, Lilly-Point precedent) and limited cash runway, the supervisory board should define an exit pathway before financing becomes urgent.
5. **Leverage 20+ IIS:** Investigator-initiated studies generate indication expansion data at minimal cost to Pentixapharm. CXCR4 imaging data from these studies directly opens new therapeutic pathways.

6. Key Takeaways

1. **Theranostic advantage:** The CXCR4 ligand as a shared scaffold for diagnosis and therapy is Pentixapharm's core differentiation — no ADC, BiTE, or CAR-T pure-play has this integrated approach.
2. **Five indications from one asset:** CNS Lymphoma, AML, Multiple Myeloma, Lymphoma, Hypertension — the pipeline breadth at minimal additional MOA risk is a structural advantage over single-asset biotechs.
3. **Diagnostic before therapy:** The PANDA Phase 3 start (H2 2026) is a regulatory milestone that de-risks the entire pipeline. Positive Phase 3 could establish CXCR4 imaging as standard — with therapy following the same path.
4. **M&A wave at the back:** 2023–2026 was the most active M&A environment in radiopharma history. Novartis (AAA \$3.9B), BMS (Rayzebio \$4.1B), Lilly (Point \$1.4B) have demonstrated that CXCR4-like platform companies can achieve billion-dollar valuations.
5. **Risk: small studies, long timeline:** 15 patients in CNS Lymphoma over 4 years and no monotherapy data are simultaneously realistic for an orphan indication and an investor communication risk. Transparent dose-escalation updates are critical.

Bottom Line: Pentixapharm occupies a unique space at the intersection of radiopharma and theranostics. The platform has the potential to define CXCR4 diagnostics and therapy across multiple indications — provided that clinical execution in CNS Lymphoma and AML delivers the first proof-of-concept data within the current cash runway.

7. Sources & Data Verification

All data points in this report have been verified against primary sources on 27 May 2026.

Clinical Trials

- NCT06132737 — [⁹⁰Y]Y-PTT in CNS Lymphoma (ClinicalTrials.gov, accessed 27 May 2026)
- NCT05351593 — Tafasitamab + Lenalidomide in CNS Lymphoma (ClinicalTrials.gov)
- NCT06827782 — Anti-CD19 CAR-NK in CNS Lymphoma (ClinicalTrials.gov)
- NCT05188872, NCT06247566, NCT06756737, NCT06635993, NCT05839483 — PentixaFor diagnostic studies (ClinicalTrials.gov)

EMA Approved Drugs

- EMA Human Medicines Register — CNS Lymphoma / PCNSL search: **0 approved drugs** specifically for PCNSL
- Condition search: "lymphoma, non-hodgkin" returns 64 results; none specific to CNS

Publications

- PMID: 39744224 — CXCR4-directed endoradiotherapy with [¹⁷⁷Lu]Pentixather + TBI in R/R AML (Theranostics, 2025)
- PMID: 34413147 — Biokinetics and Dosimetry of ¹⁷⁷Lu-Pentixather (J Nucl Med, 2022)
- PMID: 35674738 — CXCR4-targeted theranostics in oncology (Eur J Nucl Med, 2022)
- PMID: 30850502 — Side effects of Pentixather before HSCT (J Nucl Med, 2019)
- PMID: 29290814 — Dual targeting of acute leukemia by CXCR4 theranostics (Theranostics, 2018)

Company Sources

- Pentixapharm Website (pentixapharm.com) — Pipeline, Management, News — accessed 27 May 2026
- Pentixapharm News 25 Sep 2025 — PentixaTher advancement to Dose Level 4 in AML
- Pentixapharm Ad-hoc 23 Oct 2025 — Pipeline prioritisation, preclinical halt, reduced loss forecast
- Pentixapharm News 6 May 2026 — Q1 2026 Results
- Pentixapharm News 10 May 2026 — PANDA Phase 3 IND submission to FDA
- Pentixapharm News 25 Feb 2026 — FDA "Study May Proceed" for dual hemato-oncology INDs
- Pentixapharm News 5 Feb 2026 — Peer-reviewed Phase 2 PentixaFor data in Primary Aldosteronism

Financial Data

- Yahoo Finance (PTP.DE) — stock price, 52-week range, volume (accessed 27 May 2026)
- Pentixapharm Holding AG — Q1 2026 Report (6 May 2026)
- Ad-hoc announcement 6 Nov 2025 — reduced loss forecast for FY2025

Data Sources Used

- ClinicalTrials.gov — trial protocols, results, and status
- EMA Human Medicines Register — EU approval status
- PubMed — scientific literature
- Open Targets (EMBL-EBI) — drug-target relationships
- Yahoo Finance — market data
- Company website — corporate and pipeline information